Cheat sheet

Geavanceerde Computerarchitectuur (GCA)

This document lists several functions, definitions, etc., handy for the GCA lab. It is a living document, where updates are added according to the lab sessions and feedback from the students. Thanks for pointing things out.

GPU memory allocation

```
float* d_A;
cudaMalloc(&d_A, size);
```

Memory migration to / from GPU

```
float* h_A = (float*)malloc(size);
cudaMemcpy(d_A, h_A, size, cudaMemcpyHostToDevice);
cudaMemcpy(h_A, d_A, size, cudaMemcpyDeviceToHost);
```

Kernel invocation (100 blocks, with 80 threads each)

```
foo<<<100, 80>>>(d_A);
```

Thread indexing

```
int i = threadIdx.x;
```

Thread indexing

```
int i = threadIdx.x;
```

Setting to global scope (invoked by CPU or GPU)

```
__global__ void foo(d_A){...}
```

Setting to GPU scope scope (GPU)

```
__device__ float foo(d_A){...}
```

Timing (CPU)

```
#include <chrono>
const auto start = std::chrono::steady_clock::now();
...
const auto end = std::chrono::steady_clock::now();
const std::chrono::duration<double> elapsed_seconds{end - start};
```

Timing (GPU)

```
cudaEvent_t start, stop;
cudaEventCreate(&start);
cudaEventCreate(&stop);
cudaEventRecord(start);
...
cudaDeviceSynchronize();
cudaEventRecord(stop);
float milliseconds;
cudaEventElapsedTime(&milliseconds, start_cuda, stop_cuda);
```